Unit: Unit 0: Foundations of Algebra
Topic: Variables and Expressions / Order of Operations

Name: ____________

Date: __________

Advanced (Habanero)

Q1. Evaluate the expression $2 \cdot 3 + 12 \div 4$.
(A) 9
(B) 10
(C) 11
(D) 12
(E) 13

Q2. Translate the phrase 'five less than twice a number' into an algebraic expression.
(A) $2x - 5$
(B) $2x + 5$
(C) $5 - 2x$
(D) $5 + 2x$
(E) $2x + 3$

Q3. Simplify the expression $3x + 2x + 4$.
(A) $5x + 4$
(B) $5x - 4$
(C) $3x - 2$
(D) $2x + 1$
(E) $x + 3$

Q4. Evaluate the expression $x^2 + 3x - 2$ when $x = 4$.
(A) 16
(B) 20
(C) 24
(D) 28
(E) 30

Q5. Evaluate the expression $12 \div 3 + 2 \cdot 2$.
(A) 8
(B) 10
(C) 12
(D) 14
(E) 16

Q6. Translate the phrase 'the product of two numbers' into an algebraic expression.
(A) $x + y$
(B) $x - y$
(C) $x \cdot y$
(D) $x \div y$
(E) x^2

Q7. Simplify the expression $2(3x - 1) + x$.
(A) $5x - 2$
(B) $6x - 2$
(C) $7x - 2$
(D) $7x + 2$
(E) $6x + 1$

Q8. Evaluate the expression $(2x + 1) + (3x - 2)$ when $x = 2$.
(A) 7
(B) 9
(C) 11
(D) 13
(E) 15

Open Ended Questions (FRQ)

Q9. Simplify the expression $(x + 2)(x - 3)$ and evaluate it when $x = 4$.

Q10. Translate the phrase 'twelve more than the product of two numbers' into an algebraic expression and simplify it.

Chef's Tips

- Read each question carefully.
- Use the correct order of operations.
- Check your work.